

数学笔记

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Chapter 1

Symplectic Geometry

1.1 Symplectic Geometry and Hamiltonian Mechanics

Let $(Q(q, p), P(q, p))$ be a transformation from (q, p) to (Q, P) . The Jacobian matrix of this transformation is

$$J = \frac{\partial(Q, P)}{\partial(q, p)} = \begin{pmatrix} \frac{\partial Q}{\partial q} & \frac{\partial Q}{\partial p} \\ \frac{\partial P}{\partial q} & \frac{\partial P}{\partial p} \end{pmatrix} =: \begin{pmatrix} A & B \\ C & D \end{pmatrix}.$$

Theorem 1.1.1. *The Poisson bracket condition is equivalent to J^T being symplectic.*

Proof. The Poisson bracket condition is

$$\begin{aligned} \{P_k, Q_l\}_{q,p} &= \sum_{i=1}^n \left(\frac{\partial Q_l}{\partial q_i} \frac{\partial P_k}{\partial p_i} - \frac{\partial Q_l}{\partial p_i} \frac{\partial P_k}{\partial q_i} \right) = \delta_{kl}, \\ \{Q_k, Q_l\}_{q,p} &= \sum_{i=1}^n \left(\frac{\partial Q_k}{\partial q_i} \frac{\partial Q_l}{\partial p_i} - \frac{\partial Q_k}{\partial p_i} \frac{\partial Q_l}{\partial q_i} \right) = 0, \\ \{P_k, P_l\}_{q,p} &= \sum_{i=1}^n \left(\frac{\partial P_k}{\partial q_i} \frac{\partial P_l}{\partial p_i} - \frac{\partial P_k}{\partial p_i} \frac{\partial P_l}{\partial q_i} \right) = 0. \end{aligned}$$

Written in a tighter form, they are

$$\begin{aligned} \frac{\partial Q}{\partial q} \left(\frac{\partial P}{\partial p} \right)^T - \frac{\partial Q}{\partial p} \left(\frac{\partial P}{\partial q} \right)^T &= I \Rightarrow AD^T - BC^T = I, \\ \frac{\partial Q}{\partial q} \left(\frac{\partial Q}{\partial p} \right)^T - \frac{\partial Q}{\partial p} \left(\frac{\partial Q}{\partial q} \right)^T &= 0 \Rightarrow AB^T - BA^T = 0, \end{aligned}$$

$$\frac{\partial P}{\partial q} \left(\frac{\partial P}{\partial p} \right)^T - \frac{\partial P}{\partial p} \left(\frac{\partial P}{\partial q} \right)^T = 0 \Rightarrow CD^T - DC^T = 0.$$

This is equivalent to say

$$\begin{aligned} J\Omega J^T &= \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} \begin{pmatrix} A^T & C^T \\ B^T & D^T \end{pmatrix} \\ &= \begin{pmatrix} AB^T - BA^T & AD^T - BC^T \\ CB^T - DA^T & CD^T - DC^T \end{pmatrix} \\ &= \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} = \Omega. \end{aligned}$$

□

Theorem 1.1.2. *The differential form condition is equivalent to J being symplectic.*

Proof. Let $\omega_{(q,p)} = \sum_{i=1}^n dq_i \wedge dp_i$, $\omega_{(Q,P)} = \sum_{i=1}^n dQ_i \wedge dP_i$. Then

$$\begin{aligned} \omega_{(Q,P)} &= \sum_{i=1}^n \left(\sum_{k=1}^n \frac{\partial Q_i}{\partial q_k} dq_k + \frac{\partial Q_i}{\partial p_k} dp_k \right) \wedge \left(\sum_{l=1}^n \frac{\partial P_i}{\partial q_l} dq_l + \frac{\partial P_i}{\partial p_l} dp_l \right) \\ &= \sum_{i=1}^n \sum_{k,l=1}^n \left(\frac{\partial Q_i}{\partial q_k} \frac{\partial P_i}{\partial p_l} - \frac{\partial Q_i}{\partial p_l} \frac{\partial P_i}{\partial q_k} \right) dq_k \wedge dp_l \\ &\quad + \sum_{i=1}^n \sum_{k < l} \left(\frac{\partial Q_i}{\partial q_k} \frac{\partial P_i}{\partial q_l} - \frac{\partial Q_i}{\partial q_l} \frac{\partial P_i}{\partial q_k} \right) dq_k \wedge dq_l \\ &\quad + \sum_{i=1}^n \sum_{k < l} \left(\frac{\partial Q_i}{\partial p_k} \frac{\partial P_i}{\partial p_l} - \frac{\partial Q_i}{\partial p_l} \frac{\partial P_i}{\partial p_k} \right) dp_k \wedge dp_l. \end{aligned}$$

Thus $\omega_{(Q,P)} = \omega_{(q,p)}$ if and only if

$$\begin{aligned} \sum_{i=1}^n \left(\frac{\partial Q_i}{\partial q_k} \frac{\partial P_i}{\partial p_l} - \frac{\partial Q_i}{\partial p_l} \frac{\partial P_i}{\partial q_k} \right) &= \delta_{kl}, \\ \sum_{i=1}^n \left(\frac{\partial Q_i}{\partial q_k} \frac{\partial P_i}{\partial q_l} - \frac{\partial Q_i}{\partial q_l} \frac{\partial P_i}{\partial q_k} \right) &= 0, \\ \sum_{i=1}^n \left(\frac{\partial Q_i}{\partial p_k} \frac{\partial P_i}{\partial p_l} - \frac{\partial Q_i}{\partial p_l} \frac{\partial P_i}{\partial p_k} \right) &= 0. \end{aligned}$$

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Written in a tigher form, they are

$$\begin{aligned}\left(\frac{\partial Q}{\partial q}\right)^T \frac{\partial P}{\partial p} - \left(\frac{\partial P}{\partial q}\right)^T \frac{\partial Q}{\partial p} &= I \Rightarrow A^T D - C^T B = I, \\ \left(\frac{\partial Q}{\partial q}\right)^T \frac{\partial Q}{\partial p} - \left(\frac{\partial P}{\partial q}\right)^T \frac{\partial P}{\partial p} &= 0 \Rightarrow A^T B - C^T A = 0, \\ \left(\frac{\partial Q}{\partial p}\right)^T \frac{\partial P}{\partial q} - \left(\frac{\partial P}{\partial p}\right)^T \frac{\partial Q}{\partial q} &= 0 \Rightarrow B^T D - D^T B = 0.\end{aligned}$$

This is equivalent to say

$$\begin{aligned}J^T \Omega J &= \begin{pmatrix} A^T & C^T \\ B^T & D^T \end{pmatrix} \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} \\ &= \begin{pmatrix} A^T C - C^T A & A^T D - C^T B \\ B^T C - D^T A & B^T D - D^T B \end{pmatrix} \\ &= \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} = \Omega.\end{aligned}$$

□

Theorem 1.1.3. *J is symplectic if and only if J^T is symplectic.*

Proof.

$$\begin{aligned}J \text{ is symplectic} &\Leftrightarrow J^T \Omega J = \Omega \\ &\Leftrightarrow J^{-1} \Omega (J^T)^{-1} = \Omega \\ &\Leftrightarrow J \Omega J^T = \Omega \\ &\Leftrightarrow J^T \text{ is symplectic.}\end{aligned}$$

□

Those three theorems together give the following conclusion:

Theorem 1.1.4. *The Poission bracket condition is equivalent to the differential form condition.*